C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 2.01

Simultaneous Equations & Inequalities

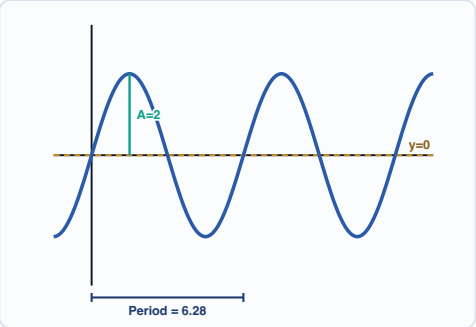
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Simultaneous Equations & Inequalities Practice Exercises

1 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



A 57 units per second

B 19 units per second

C $\frac{19}{3}$ units per second

D 21 units per second

2 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x - 2)$.

A -10

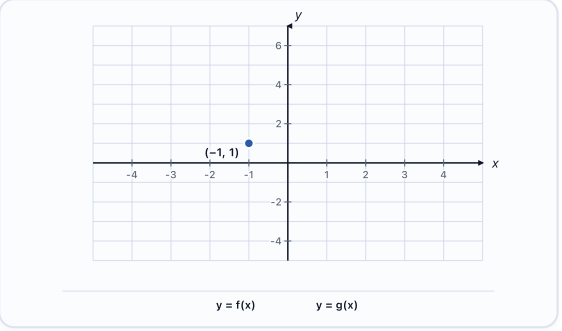
B -7

C -12

D -22

3 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 27

B 36

C 54

D 72

4 MCQ

Solve the logarithmic equation $\log_2(5x+6) - \log_2(x) = 3$.

A $x = 2$

B $x = 6$

C $x = \frac{1}{2}$

D $x = -3$

5 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_3 = 10 \times {}^{n}\mathrm{C}_2$$

A $n = 7$

B $n = 12$

C $n = 5$

D $n = 8$

6 MCQ

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{9}{2}$

D 0

7 MCQ

In $\triangle ABC$, sides of lengths 32 and 10 meet at an included angle of 60° opposite side x. Find the squared length $f = x^2$ by the Law of Cosines.

A 804

B 803

C 820

D 809

8 MCQ

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

A $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$

B $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$

C $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$

D $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

9 MCQ

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

A 3

B 12

C 6

D 14

10 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x?

A 8
B 16
C 4
D 18

11 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x?

A 5
B 22
C 10
D 20

12 MCQ

Obtain the value(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x?

A 12
B 24
C 6
D 26

13 MCQ

Work out the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x?

A 14
B 30
C 28
D 7

14 MCQ

Determine the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x?

A 16
B 8
C 32
D 34

15 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

A 9
B 38
C 18
D 36

16 MCQ

Let $f(x) = \dfrac{140x^2 + 10x - 10}{x^2 + 14}$ for $x > 0$. Find the limit at infinity $\lim_{x \to \infty} f(x)$.

A 143

B 126

C 150

D 140

17 MCQ

The quadratic graph $f(x) = 4x^2 + gx + 256$ is tangent to the horizontal axis with $g > 0$. Find the positive coefficient g.

A 64

B 68

C 72

D 54

18 MCQ

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant.\n\nGiven that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

A 6 m

B 4 m

C 5 m

D 8 m

19 MCQ

The radical function $f(x) = \sqrt{7x + 8}$ is defined for $7x + 8 \geq 0$. Find the product $7x$ when $f(x) = 8$.

A 63

B 56

C 48

D 62

20 MCQ

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

A 13

B 8

C 7

D 10

21 MCQ

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants.
Find the remainder when $P(x)$ is divided by $x - 3$.

A 12
B -150
C -30
D 60

22 MCQ

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

A 14 cm
B 8 cm
C $2\sqrt{19}$ cm
D $2\sqrt{34}$ cm

23 MCQ

Let $\mathbf{f} = \langle 12, 17 \rangle$ and $\mathbf{g} = \langle 8, 3 \rangle$ be vectors in R^2. Find the dot product $x = \mathbf{f} \cdot \mathbf{g}$.

A 172
B 130
C 147
D 159

24 MCQ

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2.
During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm .

A 50π
B $\frac{36\pi}{5}$
C 20π
D 11π

25 MCQ

Find the set of values of the constant k for which the equation

$$|x^2 - 4| = 2x + k$$

has exactly four distinct real solutions.

A $4 \le k \le 5$
B $4 < k < 5$
C $-4 < k < 4$
D $k > 5$